Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

# SDC Hackathon

# Technical Tools Declaration

**Hosted by Xiamen University Malaysia (XMUM) AI Club**

**Submission Deadline: [Insert date]**

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

***Instagram:*** @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

# 1. TEAM AND PROJECT OVERVIEW

**Team Name**

4bit

**Project / Solution Name**

Skilytics

**Name (Leader)**

Muhammad Ameer Yusoff Bin Azeri

**Email & Phone (Leader)**

ameeryusoff1357@gmail.com / +60 11-5723 6830

**University Affiliation(s)**

Universiti Teknologi Malaysia

**Live Prototype URL**

https://skilytics.vercel.app

**Code Repository URL (public / access granted to judges)**

https://github.com/ameer-yusoff/SDC2026_Hackathon_4bit_Skilytics

# 2. ARCHITECTURE & TECHNOLOGY STACK

## 2.1. System Architecture Overview

*Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.*

***[Maximum 200 words]***

- User Interface (Frontend): A React single-page application built with Vite (src/artifacts/reskill-ai) that captures user interactions, profile configurations, and RIASEC assessment responses.
- Backend API: A Node.js Express server that processes client REST requests, handling routing, authentication, and core evaluation logic.
- AI Model / API: Interacts with AI agent services (src/artifacts/api-server/src/lib/llm.ts) to process CVs, evaluate skill gaps, and generate AI-driven career recommendations.
- Database Layer: Managed via Drizzle ORM (src/lib/db) connected to a Neon Serverless PostgreSQL database, storing user profiles, skill matrices, and course catalog items.
- Cloud Hosting: Cloud infrastructure deployed via Render (render.yaml) for backend deployment and Vercel (vercel.json) for frontend hosting.

# 2. ARCHITECTURE & TECHNOLOGY STACK

## 2.2. Languages, Frameworks & Libraries

*List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.*

| Name & Version | Purpose in Project | License |
|---|---|---|
| TypeScript (v5.x) | Primary programming language used across both frontend and backend for type-safe application development. | Apache 2.0 |
| Node.js (v20.x) | Runtime environment powering the backend API server, automated build scripts, and local developer environment. | MIT |
| Python (v3.10+) | Scripting language used for model testing, sending structured inference requests to RunPod, and data validation. | PSF License |
| React (v18.x) | Frontend library for building the interactive user interface, assessment workflows, and dashboard components. | MIT |
| Vite (v5.x) | Frontend tooling and development server providing fast bundling and hot-module replacement. | MIT |
| Tailwind CSS (v3.x) | Utility-first CSS framework used for styling responsive, modern user interface layouts. | MIT |

| Name & Version | Purpose in Project | License |
| --- | --- | --- |
| Radix UI / Shadcn UI (Latest) | Unstyled, accessible component primitives used to construct dialogs, tabs, progress bars, and forms. | MIT |
| Lucide React (v0.x) | Icon library providing consistent vector icons across the platform interface. | ISC |
| TanStack React Query (v5.x) | Asynchronous state management library handling API data caching, background fetching, and server state synchronization. | MIT |
| Express.js (v4.x) | Minimalist backend web framework routing API requests for CV processing, user profiling, and market data queries. | MIT |
| Zod (v3.x) | Schema declaration and validation library enforcing structured, type-safe inputs and outputs for API routes and AI payloads. | MIT |
| Orval (v7.x) | OpenAPI code generator creating type-safe TypeScript React query hooks and fetch clients directly from OpenAPI specs. | MIT |
| Drizzle ORM (v0.x) | TypeScript Object-Relational Mapper used for database schema declaration, migrations, and type-safe database queries. | Apache 2.0 |
| Neon Serverless SDK (@neondatabase/serverless) | Serverless PostgreSQL driver connecting the backend to cloud Neon database instances via WebSockets/HTTP. | Apache 2.0 / MIT |
| Google Generative AI SDK (@google/genai) | Client SDK for interacting with the Gemini API to execute resume parsing, interest extraction, and career reasoning. | Apache 2.0 |

| Name & Version | Purpose in Project | License |
|---|---|---|
| Tesseract.js (v7.x) | Primary OCR engine to extract text from image/scanned CV files (scanned images like JPEG/PNG). | Apache 2.0 |
| PDF.js (pdfjs-dist & unpdf) (v4.x / v1.x) | Extracts text, structure, and metadata from PDF-formatted CV files. | Apache 2.0 |
| Recharts (v2.x) | Generates interactive data visualizations, radar charts (6 competency scores), and analytics charts in the dashboard | MIT |
| Framer Motion (v12.x) | Handles micro-animations, score card transitions, and smooth UI interactivity. | MIT |
| Wouter (v3.x) | Client-side routing library used for frontend page navigation (not React Router). | MIT |
| node-postgres (pg) (v8.x) | The actual driver (driver) that connects the Node.js backend directly to the Neon Postgres database. | MIT |
| Pino & Pino-HTTP (v9.x) | High-performance logging system for backend error monitoring and request auditing. | MIT |

# 3. APIs & CLOUD SERVICES

## 3.1. External APIs

*List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.*

| API and Provider | Purpose in Project | Access Tier (free / paid / sponsor-provided) |
|---|---|---|
| Google Gemini API | GenAI reasoning engine for CV parsing, RIASEC score calculation, skill gap analysis, and personalized roadmap generation. | Free |
| RunPod Serverless Inference API | REST API endpoint used by the backend to trigger serverless inference tasks on custom open-source models (Qwen / OpenJev). | Sponsor-provided |
| OpenDOSM API | Live national labor force statistics feed providing monthly employment data and local job market benchmarks. | Free |
| Anthropic Economic Index Feed | Research data feed providing country-level AI usage metrics and task-level AI exposure ratings by SOC occupation code. | Free |
| O*NET Web Services API | Standardized occupational database API providing official job taxonomy, skill definitions, and RIASEC job mappings. | Free |
| Microsoft Learn Catalog API | Ingests verified public course modules, duration estimates, and certification pathways. | Free |
| freeCodeCamp Content API | Curriculum feed supplying structured software development learning tracks and free certification units. | Free |

| Name & Version | Purpose in Project | Access Tier (free / paid / sponsor-provided) |
| --- | --- | --- |
| GitHub REST API | Content API retrieving practical coding exercises, repository templates, and technical hands-on tasks. | Free |
| Hugging Face Hub API | Content and model registry API providing machine learning course modules and open tutorial resources. | Free |

# 3. APIs & CLOUD SERVICES

## 3.2. Cloud Platforms & Deployment

*List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.*

| Service & Provider | Purpose in Project | Plan / Region |
|---|---|---|
| RunPod | Serverless cloud GPU compute hosting Docker containers for open-source decision models (Qwen / OpenJev) with automatic scale-to-zero. | Sponsor-provided |
| Neon | Serverless PostgreSQL cloud database managing user profiles, assessment submissions, cohort statistics, and market cache. | Free |
| Render | Cloud application platform hosting and running the backend Express API server container (render.yaml). | Free |
| Vercel | Edge hosting platform providing continuous deployment (CI/CD) and global CDN for the React/Vite frontend (vercel.json). | Free |
| Replit | Cloud IDE and staging environment used for interactive prototyping, testing, and multi-agent development (.replit). | Sponsor-provided |
| GitHub | Cloud Git repository hosting, version control, source code management, and automated webhooks. | Free |
| | | |

# 4. AI TOOLS, MODELS & DATASETS

## 4.1. AI Models Used in the Solution

*List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.*

| Model & Version | Source / Provider | Usage (pre-trained / fine-tuned / self-trained) | License / Terms |
|---|---|---|---|
| Gemini-flash-lite 3.1 | Google AI Studio | Pre-trained | Google APIs Terms of Service |
| Qwen 3.5-27b<br>Qwen3.5-9b | Alibaba Cloud | Pre-trained | Apache 2.0 / Qwen Open License |
| OpenJev Decision Model | Open-source Community | Pre-trained | Apache 2.0 |
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# 4. AI TOOLS, MODELS & DATASETS

## 4.2. Datasets

*List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.*

| Dataset Name | Source / URL | Type (public / private / self-collected / synthetic) | License | Personal Data? (Y/N) |
|---|---|---|---|---|
| OpenDOSM Monthly Labour Force | https://data.gov.my/data-catalogue/lfs_month | Public | CC BY 4.0 | N |
| AI Usage by Country (Anthropic Economic Index) | https://huggingface.co/datasets/Anthropic/EconomicIndex | Public | CC BY 4.0 | N |
| AI Exposure by Occupation (Anthropic Economic Index) | https://huggingface.co/datasets/Anthropic/EconomicIndex/tree/main/labor_market_impacts | Public | CC BY 4.0 | N |
| O*NET Occupation References (v31.0) | https://www.onetcenter.org/database.html | Public | CC BY 4.0 | N |
| Microsoft Learn Public Course Catalogue | https://learn.microsoft.com/api/catalog/?locale=en-us&type=learningPaths | Public | Microsoft Terms of Use | N |
| freeCodeCamp Public Curriculum | https://github.com/freeCodeCamp/freeCodeCamp/blob/main/curriculum/structure/curriculum.json | Public | CC BY-SA 4.0 / BSD 3-Clause | N |
| GitHub Skills Exercises | https://api.github.com/orgs/skills/repos?per_page=100&type=public | Public | MIT / GitHub Terms | N |
| Hugging Face Learn Catalogue | https://huggingface.co/learn | Public | Open Access / CC BY 4.0 | N |

| Dataset Name | Source / URL | Type (public / private / self-collected / synthetic) | License | Personal Data? (Y/N) |
|---|---|---|---|---|
| Holland Code (RIASEC) Question Set | https://www.onetcenter.org/dl_files/Mini-IP.pdf | Public | Public Domain | N |
| HubSpot Academy Public Catalogue | https://academy.hubspot.com/courses | Public | HubSpot Terms of Service | N |
| DOSM JMI Q4 2025 · Selected Occupation Snapshot | https://blog.gdeltproject.org/gdelt-doc-2-0-api-debuts/ | Public | CC BY 4.0 / Open Data | N |
| Google News Public Career RSS | https://news.google.com/rss/search?q=%28upskilling+OR+reskilling+OR+%22workforce+skills%22+OR+%22AI+workplace%22+OR+%22hiring+demand%22%29+Malaysia+when%3A30d&hl=en-MY&gl=MY&ceid=MY%3Aen | Public | Google Terms of Service | N |
| BLS US Career Growth Reference (2025–2035) | https://www.bls.gov/emp/tables/occupational-projections-and-characteristics.htm | Public | Public Domain (U.S. Govt) | N |
| O*NET Content Model · Skill References | https://www.onetcenter.org/content.html | Public | CC BY 4.0 | N |
| United Nations · SDG Framework | https://sdgs.un.org/goals | Public | UN Open License / CC BY 3.0 IGO | N |
| OpenLearn · Reviewed Public Courses | https://www.open.edu/openlearn/ | Public | CC BY-NC-SA 4.0 / Open University Terms | N |

| Dataset Name | Source / URL | Type (public / private / self-collected / synthetic) | License | Personal Data? (Y/N) |
|---|---|---|---|---|
| Himalayas · Public Vacancies | https://www.onetcenter.org/dl_files/Mini-IP.pdf | Public | Himalayas Terms of Use / Keyless AP | N |
| Remotive · Public Remote Vacancies | https://academy.hubspot.com/courses | Public | Remotive API Terms / Keyless API | N |
| RemoteOK · Public Job Feed | https://blog.gdeltproject.org/gdelt-doc-2-0-api-debuts/ | Public | RemoteOK Public Terms / Keyless API | N |

# 4. AI TOOLS, MODELS & DATASETS

## 4.3. AI-Assisted Development Tools

*Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.*

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

| Tool & Version | Used For | Extent of Use (minor / moderate / major) |
|---|---|---|
| Antigravity | Core agentic code generation, architecture planning, and debugging multi-agent workflow pipelines. | Major |
| OpenAI Codex | TypeScript/React code completion, helper function drafting, schema-bound logic, and unit test generation. | Moderate |
| Replit | Full-stack project scaffolding, rapid UI mockup prototyping, sandbox testing, and server environment setup. | Minor |
| Google Gemini | Prototyping multi-agent system prompts, designing structured Zod schemas, generating multi-language dictionaries (EN/BM/ZH), and technical documentation. | Major |

# 4. AI TOOLS, MODELS & DATASETS

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

***[Maximum 100 words]***

We treated all AI-generated code and model responses as rough drafts that required human checks. Our team manually reviewed and tested every piece of code before merging it into GitHub.

For AI model outputs, we enforced strict Zod schema validation to stop errors and hallucinations before data reached the frontend. We also ran automated unit tests using mock CV profiles to check parsing accuracy and course recommendations. Ultimately, our team took full ownership by testing the pipeline end-to-end to ensure all career guidance was accurate, safe, and reliable.

# 5. SPONSOR-PROVIDED RESOURCES

*Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.*

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

| Sponsor | Resource (API / credits / platform / data) | How It Was Used in the Project |
|---|---|---|
| Replit | Cloud IDE platform, development workspace, and Replit Agent tools | Used to set up the project workspace, build and test full-stack components, and prototype interactive UI flows. |
| RunPod | Cloud GPU credits and serverless container endpoints (H100 GPUs). | Used to test open-source decision models (Qwen), configure scale-to-zero serverless workers, and run structured inference via Python and cURL. |
| | | |
| | | |

# 5. SPONSOR-PROVIDED RESOURCES

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

***[Maximum 100 words]***

The sponsor resources were very helpful for fast prototyping, but they could be replaced if necessary.

RunPod allowed us to test open-source AI models on cloud GPUs without needing expensive local hardware, and Replit made it easy for our team to build and test our code together in a shared cloud workspace.

However, our core features do not depend exclusively on them. Because our platform uses standard web frameworks and flexible containerized models, RunPod could be replaced by other cloud GPU providers or commercial AI APIs, and Replit could be replaced by regular tools like VS Code.

# 6. SECURITY, PRIVACY & RESPONSIBLE USE

## 6.1. Data Handling & Privacy

*What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.*

***[Maximum 150 words]***

Skilytics collects basic account information (name and email), uploaded resumes (education, work experience, and skills), and answers from the career interest quiz. Resume text and quiz answers are sent to AI services (Google Gemini API and RunPod) strictly to extract skills, evaluate career paths, and create personalized reskilling roadmaps. We never sell or share user data with advertisers.

All data is stored in a secure cloud database (Neon Postgres) using encrypted connections (HTTPS/SSL). System credentials and API keys remain protected on our backend server so sensitive data is never exposed to the public, and our AI service providers do not use submitted data to train their public models.

Personal data is kept only while the user's account is active to display their career progress. Users can update or remove their profile at any time, which permanently removes their stored records.

# 6. SECURITY, PRIVACY & RESPONSIBLE USE

## 6.2. Known Limitations & Risks

*Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.*

***[Maximum 150 words]***

- CV Parsing Inaccuracies: AI models can misinterpret complex resume formatting or extract skills not actually present in a CV.
- Mitigation: We give users an interactive review screen to verify and edit parsed skills before processing, and we validate extracted terms against verified O*NET skill lists.

- API Downtime & Rate Limits: External services can time out.
- Mitigation: We cache frequent queries and store fallback data snapshots.

- AI Hallucinations: AI can invent courses or career facts. Mitigation: We use Zod schemas to block malformed outputs and only recommend verified courses from our connected catalogues.

- Provider Dependency: Relying on one vendor risks system failure.
- Mitigation: Our architecture supports switching between open-source models (Qwen) on RunPod and Google Gemini

# 7. COMPLIANCE & READINESS

*Tick all boxes that apply before submitting.*

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

# 8. DECLARATION

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature]



[Date]

24/10/2026